

Bayesian Techniques in Macroeconomics

Paris-Dauphine & PSL Universities
Master - Quantitative Economics, 2nd year

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CHAPTER 1

Modern Business Cycle Theory

Content of the Handout

HANDOUT OBJECTIVES

- Understanding the concept of a business cycle;
- Extracting the business cycle component from time series;
- Solving a simple rational expectation model;
- Linearizing a system of equations;

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http://vermandel.fr/QEA_NK.

1. What is a business cycle?

1.1 A brief history

Back in the 1930s, the emergence of national accounting allowed economists to get an accurate measure of aggregate production from the demand, supply and income sides. National accounts provide a complete and consistent set of techniques to measure a country's economic activity. Given these new measures of economic activity, the study of business cycle quickly emerged. As explained by ?, the pioneer of the concept of business cycle can be traced back to ? work on the interpretation of gold, prices and wages during the US civil war in his treatise on *Business Cycles*. He had provided the first quantitative study of business cycles. His pioneered work provided the basis of modern business cycle analysis in ? by defining a business cycle as “a *type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; in duration, business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar characteristics with amplitudes approximating their own.*”

The contributions of Mitchell provide the first dichotomy in economics by separating long versus short term in macroeconomic analysis. In particular, ? provide a first definition of the duration of a business cycle, the latter lies between one to twelve year. This definition of business cycle will change over time, modern business cycle analysis determine the frequency of a business cycle as a fluctuation lying between 2 to 32 quarters, so a duration shorter than ?. Every component of macroeconomic time series with a frequency lower than 32 years is defined as the trend of the series.

A clear topology of an economic cycle is a first order aspect of macroeconomic analysis, in particular for policymakers. Consider a government facing an adverse event (financial turmoil, pandemic, war episode, etc.), dating the recession, gauging its magnitude and duration are all key to determine the size of any fiscal/monetary stimulus to dampen the crisis.² During the last century, economic associations – exemplified by the NBER and CEPR – were established in order to provide start and end dates of recessions for the US and Europe, respectively. Therefore, many efforts have been done by economists to provide a clear typology of the economic cycle by disentangling periods of expansion, crisis, recession and recovery in macroeconomic time series.

Prior the 40s, most of the economic cycle analysis was about determining an anatomy its periodicity. You may have heard during your undergraduate studies about the Juglar, Kuznets and Kondratiev cycles. These cycle analysis aims at explaining how long an innovation, an investment, or any other spending may create a long lasting cycle (i.e. dating the duration of an expansion cycle followed by a recession). Following the contribution of ?, the cycle analysis shifted to disentangling any time series into a cyclical vs trend components. This change in the cycle analysis also resulted in some theoretical interpretations of the cycle through the establishment of a short run and long run macroeconomics (fluctuations vs growth). These materialize through new theories of economic growth (e.g. ?, ?) and theories of fluctuations (e.g. ?, ?). In this lecture, we will only work on the business cycle theory which emerged in the 70s-80s.

²For instance in the European Union, there is an ongoing debate since Maastricht Treaty about fiscal rules and public deficits. In order to provide a government budget balance, the EU may penalize any EU country exhibiting too important structural deficits. The critical aspect lies in the definition of the 'structural' nature deficit, which typically relies on business cycle measures. A change in the computation of the cyclical budget deficit may change the stance of the EU commission about national fiscal rules.

1.2 Measuring business cycles

However, the frontier separating short run and long run analysis is however not as clear as what one would expect. The emergence of econometrics in the second half of the 20th century had also been accompanied by new statistical methods that aim at separating low and high components in economic time series. Any macroeconomist working in a time series face the following problem:

$$y_t = \tau_t + \psi_t, \quad (1)$$

where τ_t is the secular and ψ_t the cyclical component of time series y_t . How can we compute τ_t and ψ_t ? Unfortunately, there is no unique method to disentangle τ_t and ψ_t . Therefore, the measure of business cycles is still open equation. In this section, we will discuss three different methods to extract the business cycle component from the data.

A first preliminary step prior working on real data is to apply logs. Why do we use logs? Because GDP data exhibits an exponential growth component. To illustrate this aspect, let us assume for now that economic growth is following an exponential growth process (a very common assumption). Formally, Let Y_t denote the GDP that follows the following data-generating process:

$$Y_t = Y_0 (1 + \gamma)^t \exp(\varepsilon_t). \quad (2)$$

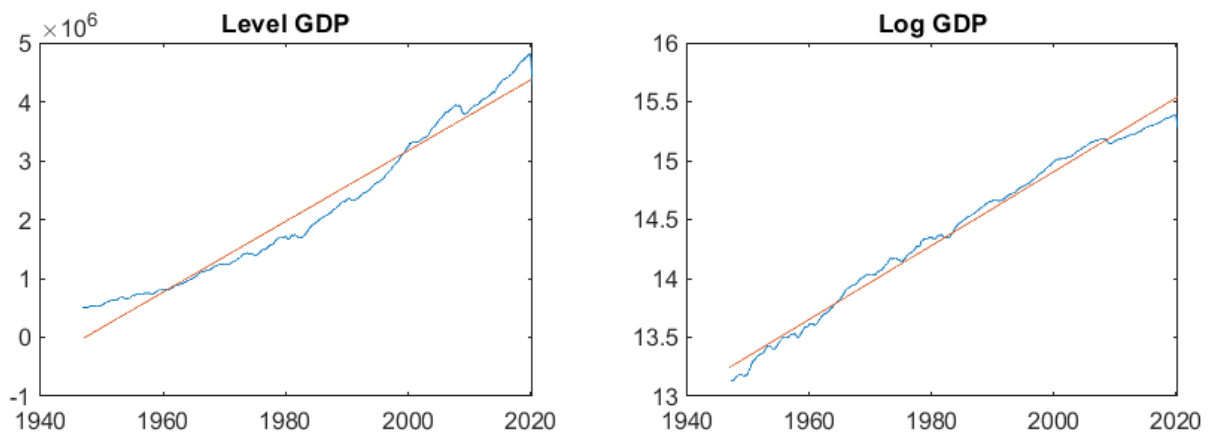
In this expression, $(1 + \gamma)^t$ is the (exponential) trend, ε_t is the fluctuation term and Y_0 is the initial observation of output. In this expression, the interpretation of γ is simply the quarterly percentage growth rate. Applying logs to this expression to linearize expression (??):

$$\log(Y_t) = \log(Y_0) + \log(1 + \gamma)t + \varepsilon_t \quad (3)$$

$$\simeq \underbrace{y_0 + \gamma \cdot t}_{\tau_t} + \underbrace{\varepsilon_t}_{\psi_t}. \quad (4)$$

Note that parameters y_0 and γ be estimated by OLS. From these estimated parameters, the trend can be computed and removed from level output to obtain the fluctuation term ψ_t .

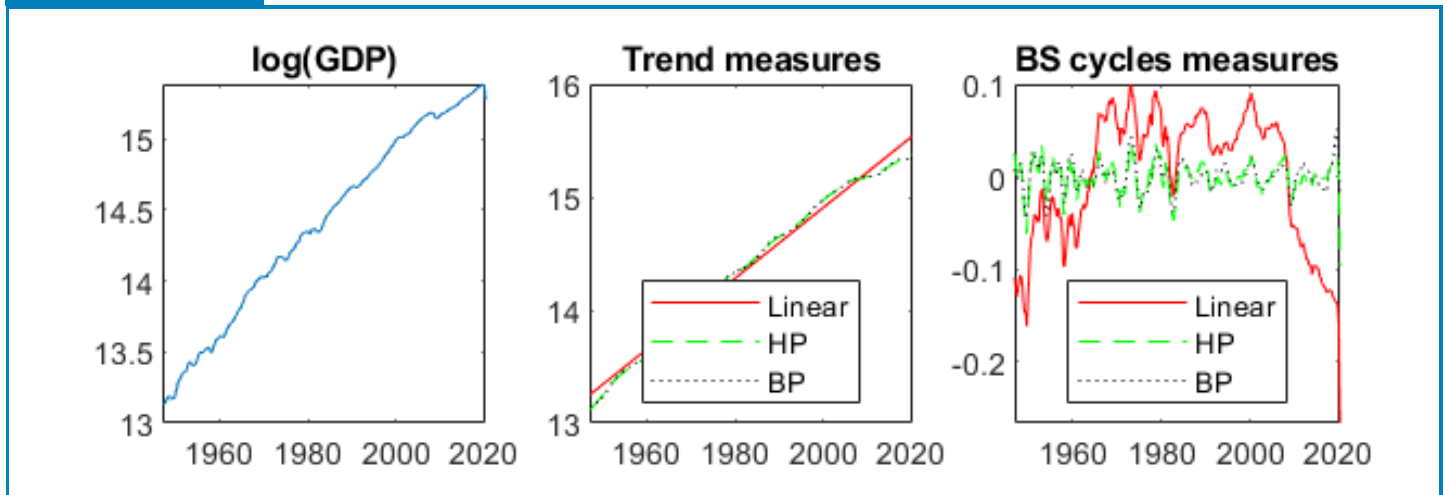
MATLAB box 1 `logGDP.m`



We apply three filtering methods on the log of GDP to illustrate some alternative measures of business cycles:

1. **A linear trend:** in a nutshell, we simply use OLS regression and estimate the slope of growth. This method assumes that the data-generating process is the same as in ???. This method allows to quickly extract the business cycle component without sweeping information in the data. However, if output permanently drifts following a structural break, then the measure of business cycle may be biased (e.g. following the great recession in 2007, the US has never reached the pre-crisis trend, which makes the business cycle component to be permanently negative after 2007). Given the possible existence of some structural change in economic growth, it may be useful to consider a time-varying trend.
2. **The HP filter:** the statistical method of ? extracts the cyclical component of a time series by separating long vs short-term fluctuations. Note that this method is two-sided, which means that to be computed in period t , the filter requires both past and futures values. So end of sample estimations are typically biased. In addition, the filter sweep away some informations in the data that may be relevant for the analysis. Therefore, this method faces many critiques summarized by ?.
3. **The band-pass filter:** a band-pass filter is a device that passes frequencies within a certain range and rejects (attenuates) frequencies outside that range. In economics, the band pass filter can be used to isolate the component of a time series that lies within a particular band of frequencies to separate the business cycle, the medium term cycle and the long term cycle (i.e. the trend). This allows for instance to examine if an economic relation (e.g. money-inflation) remains significant for different frequencies of the cycle. ? and ? provide the two most famous applications of the bandpass filter to business cycle analysis.

MATLAB box 2 BSmeasures.m



1.3 The analysis of business cycles

? provides a nice description of modern business cycle analysis: “*Fluctuations in economic activity are seen as the result of small, white noise shocks that affect the economy through a complex dynamic propagation system.*” Pioneered by ?, theoretical interpretations of economic fluctuations (i.e. the term ε_t in ??) rely on system response analysis following the realization of some shocks. The dominant view is that even large fluctuations result from small shocks.³ Particular economic fluctuations can be ascribed to particularly large shocks followed by periods during which the economy returns to equilibrium. This interpretation

³Some more recent applications to financial turmoils, such as ? or ?, show that small shock can trigger large depressions.

contrasts with the “old” Keynesian approach of economic fluctuations: the latter interpret fluctuations as imbalances in money or labor markets.

According to ?, the business cycle exhibits some regularities stable over time:

1. The main regularity is at a sectoral level: output movement across sector are strongly serially correlated. So if one sector experiences one boom, it's very likely that all other sector experience also experience a boom too.
2. Production of durable goods is more volatile than non-durable goods.
3. Price are procyclical.
4. Short term interest rates are procyclical.
5. Monetary aggregates and velocity measures are procyclical.

Given these observations, ? argued that “all business cycle are alike”. This assertion has important implications for theorists, as it implies that these business cycle regularities could be all interpreted from a single general theory of fluctuations. In the following years, Lucas has provided several contributions to the foundation of a theory of business cycles. The main ingredients are a general equilibrium with infinitely-living and representative agents that form their expectations rationally. This handout provides the basis of business cycle theory pioneered by Bob Lucas.

2. An illustration of a business cycle model through the consumption-saving problem

According to ?, a theory of real business cycle aims at providing any point in time de value of output. In particular, it provides some theoretical interpretations of the empirical relations across macroeconomic time series. In this section, we present the consumption-saving problem to examine how changes in (expected) consumption affect the equilibrium real rate.

2.1 A toy model of consumption-saving

Consider an simplified version of the ?'s asset pricing model. This model is characterized by discrete-time and an infinite horizon economy populated by *households*, which are infinitely lived and of measure one. This is an endowment economy, so the production is assumed to be fully exogenous and driven by an exogenous shock:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \text{ with } \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \quad (5)$$

The production in this economy stems from the presence of infinitively-lived trees, output is the fruit that falls from trees. Therefore, we interpret ε_t as the weather that disrupt the quantity of fruits that the tree can provide (i.e., for a low realization of ε_t , the tree provides a lower quantity of fruits). The weather exhibits some persistence, so a low realization of the weather today is likely to affect the quantity of fruits in the next period by a degree ρ . To illustrate this persistence, a storm may damage a branch, the latter requires few periods to heal and produce fruit as usual. In this model, the only source of uncertainty is about current and future values of ε_t , as the household ignore how much of fruits she will have. This uncertainty can be addressed by the purchased of some financial claims that we discuss hereafter.

Regarding the preferences, each household maximizes her lifetime utility, measured as the current and future discounted streams of utility:

$$\max_{\{c_t, b_t\}} \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s u(c_{t+s}), \quad (6)$$

where β is the discount factor.⁴ We impose a logarithm for the utility function $u(c_t) = \log(c_t)$. Expression $\mathbb{E}_t$ denotes the expectation operator. In a rational expectation equilibrium, agents can form expectations about the future states of the economy at no cost. However in a stochastic environment, the expectation of $\mathbb{E}_t \{X_{t+1}\}$ is assessed with all information available in t to agents, meaning that any shock that will occurs between t and $t + 1$ will not be anticipated: a gap between the expectation of $\mathbb{E}_t \{X_{t+1}\}$ and its realization X_{t+1} is observed, thus creating a source of fluctuations.

Regarding the financial aspect of the model, we depart from ? to introduce a (risk free) financial claim in the model. Given the uncertainty stemming from the weather, the household may be interested in purchasing a financial claim to hedge against low realizations of the weather. Put differently, the household may choose to consume all the fruits today, or postpone consumption to the next period by depositing the fruits to a bank. The bank is perfectly competitive, and provide (fruit) deposit services and provide an interest rate r_t for each unit of fruit deposited between t and $t + 1$. The fruit is the numeraire, so the interest rate r_t is directly interpreted in units of fruits.

⁴For $\beta = 1$, the household is perfectly patient and does not discount the future, *i.e.* consuming now or later provides exactly the same utility ($\beta^0 u(c) = \beta^1 u(c) = \beta^2 u(c)$). For any value of $\beta < 1$, the household is impatient, the lower β is, the more impatient the household is ($\beta^0 \mathcal{U}(c) > \beta^1 \mathcal{U}(c) > \beta^2 \mathcal{U}(c)$). Microeconomic evidence suggests that households are impatient, so the discount factor should lie between 0 and 1 (usually close to one).

2.2 A discussion of the first order conditions

So each period, the households has to possibility to consume c_t units of fruits or deposit b_t to the bank. The budget constraint faced by the household at any point in time is given by:

$$c_t + b_t \leq b_{t-1}r_{t-1} + y_t, \quad (7)$$

where the left-hand side of the equation gathers the spending of the household, while the right-hand side gather the income from deposits and the production from trees. Note that when the household takes it deposit decisions, she knows the interest rate, r_{t-1} , so there is no uncertainty on the return the household may get from depositing fruits in $t + 1$.

The Lagrangian problem solved by the household is given by:

$$\mathcal{L}_t = \sum_{s=0}^{\infty} \beta^s \left[\log(c_{t+s}) - \lambda_{t+s} (c_{t+s} + b_{t+s} - b_{t-1+s}r_{t-1+s} - y_{t+s}) \right], \quad (8)$$

where λ_t denotes the Lagrangian multiplier over the budget constraint. Two important assumptions are implicitly set on this Lagrangian problem: (i) the budget constraint is also discounted, so the constraint is eternally binding over time (i.e., it binds today and in the future); (ii) the budget constraint binds to equality, so the income is fully spent in ??.⁵

Maximizing ?? in c_t and b_t yield the following first order conditions:

$$(c_t) : \lambda_t = \frac{1}{c_t} \quad (9)$$

$$(b_t) : \lambda_t = \beta E_t \{ \lambda_{t+1} \} r_t \quad (10)$$

Note that if the household consumes and saves by adopting these two equations, she will get the highest feasible level of lifetime utility over the business cycles. We provide an interpretation of these two equations in the following subsection.

2.2 A discussion of the first order conditions

The two first order conditions (??-??) are one of the main ingredient of real business cycle theory: the [Euler equation](#).

o Marginal utility of consumption (MUC).

In equilibrium, the Lagrangian multiplier over the budget constraint is equal to the marginal utility of consumption: $\lambda_t = u'(c_t)$. The functional form of the utility function $u(c_t)$ critically matters to explain the saving-consumption nexus. So this section provides some insights about the interpretation of the utility function.

Consider an exogenous consumption cycle as defined in ??, what are the implications of this cycle in terms of utility function $u(c_t)$ and marginal utility of consumption λ_t for the household? ?? reports some illustrative simulations of consumption, and their corresponding streams of utility and marginal utility.

When consumption is high in period 3, the individual utility also increases with the level of consumption, so there is a perfect co-movement between utility and consumption. In contrast, the marginal utility exhibits a different pattern of fluctuations with respect to consumption. When consumption is high, the willingness to consume is low. Why? Because of **diminishing marginal utility**, the consumption of each

⁵Note that we can check whether the constraint always binds to equality by tracking the non-negativity of the Lagrangian multiplier.

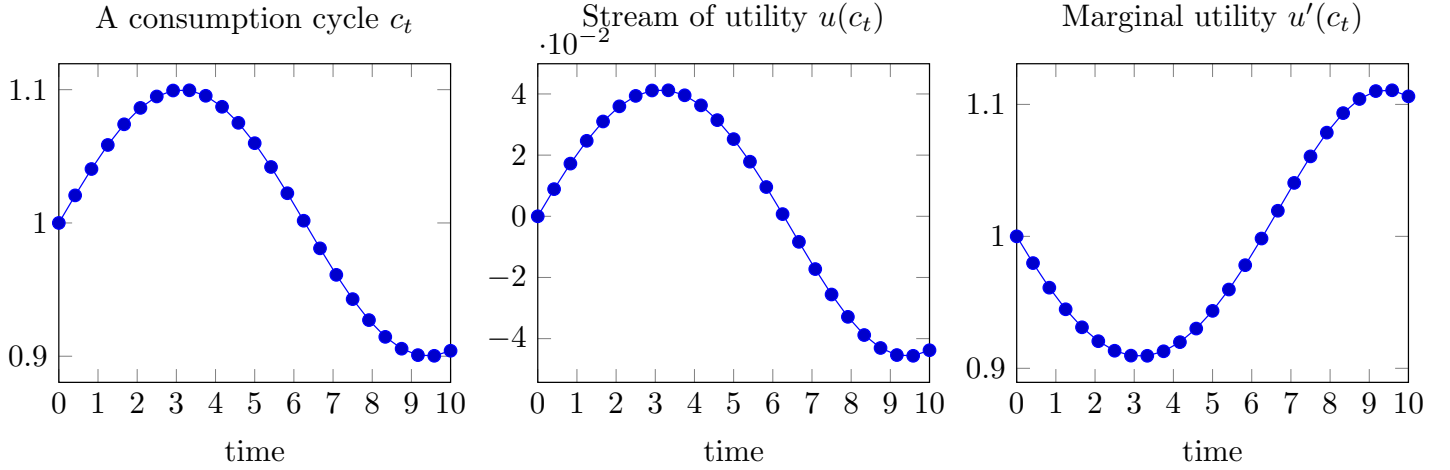


Figure 1: Consumption, utility and marginal utility cycles under logarithmic utility

additional unit of a good gives less extra utility than the consumption of the previous unit. Think about a cyclist running the Tour de France who is thirsty following some physical effort to run up a hill. At first, he's really willing to drink water (λ_t is high), so the first glass of water provides a lot of utility. For the second glass, he becomes less thirsty so the consumption of the second glass does provide utility but not as much as the previous one. This feature of the marginal utility explains the negative relation between consumption and the willingness to consume. When consumption is low, the utility from consuming an additional unit of goods is higher than when consumption is high.

o Intertemporal Marginal Rate of Substitution (IMRS).

The second core key ingredient of modern business cycle theory is the Euler equation. Consider optimal condition on bonds:

$$u'(c_t) = \beta E_t \{u'(c_{t+1})\} r_t. \quad (11)$$

The left hand side of this equation provides the utility loss of saving today (priced in terms of present marginal utility), and the right hand side is the utility gain at the next period. As long as this equation holds to equality, there is a no-arbitrage condition that makes the household equally well off between consuming today or tomorrow. Note that the equilibrium interest rate plays here a key role as it directly determines how households are weighting consumption today versus tomorrow: if the equilibrium rate rises, households are more likely to postpone consumption and to consume more at the next period. Therefore, the key ingredient here is the pricing scheme that affects the trade-off between consuming now and tomorrow. The intertemporal marginal rate of substitution of future for present consumption is given by $u'(c_t)/E_t \{u'(c_{t+1})\}$.

The intertemporal analysis of consumption is actually exactly the same as in the standard consumer problem describing the maximization of the utility by determining the optimal quantity of two types of goods under a budget constraint. The difference here is that instead of having two different goods, the trade-off now applies between consuming today and tomorrow. The budget constraint becomes intertemporal and shapes the dynamics of the interest rate.

?? reports the joint-determination of the optimal level of financial assets and consumption. In the left panel, we show the consumption possibilities (c_t, c_{t+1}) which are offered by financial markets in the economy (and implicitly here by the production capacities of the economy which are exogenous). The slope of this curve, $-\beta E_t \{r_{t+1}\}$, is decreasing because current and future consumption are substitutes.

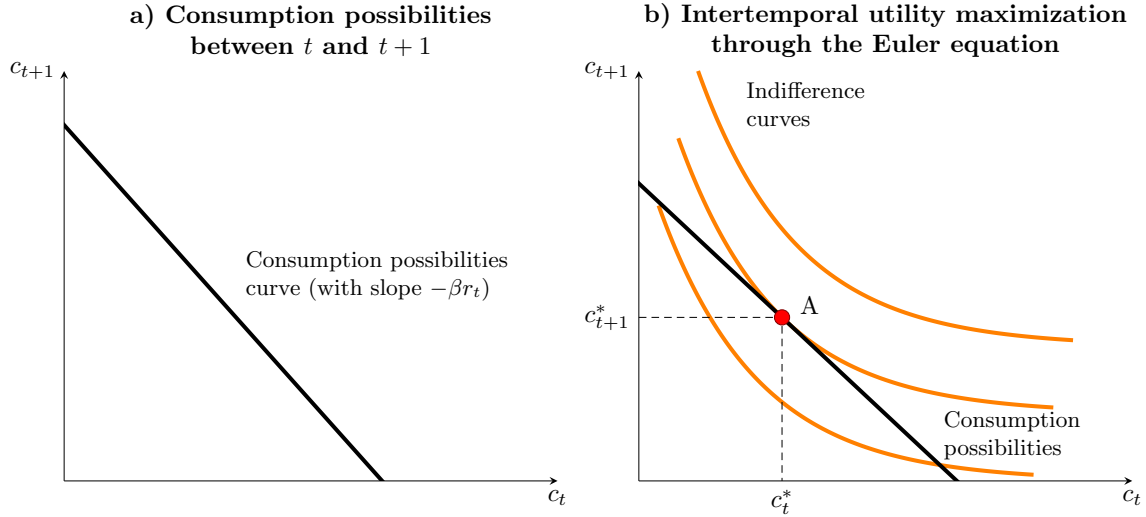


Figure 2: Optimization scheme for the representative household

In the right panel of ??, we introduce the household's preference with respect to consumption. These preferences materialize in a graphical representation as indifference curves. These curves are different pairs (c_t, c_{t+1}) that provides exactly the same utility so that an individual is equally well off in any point of the indifference curve. Given the real rate on deposits, the household determines the levels of present and future consumption (c_t^*, c_{t+1}^*) maximizing her utility given the consumption possibilities offered by financial markets. In equilibrium A, the household find the unique possible path of consumption maximizing her utility intertemporally and is called *the consumption point*. ?? is actually the (unique) consumption point that provides the highest level of lifetime utility for a given budget constraint.

2.3 General Equilibrium

Finally, we need to set one condition over deposits to close the model. In financial autarky (*i.e.*, with no trading partners), the general equilibrium is jointly determined by the equilibrium on financial and goods markets, as the optimal condition of saving determine the amount of consumption. So clearing one market also clears the others.

In absence of external borrowing/lending, at a macroeconomic level, the following condition on domestic saving emerges:

$$b_t = 0. \quad (12)$$

Note that since households populating our economy are all symmetric, the agent specific saving is also zero in equilibrium. Given this relation, the budget constraint in ?? becomes:

$$c_t = y_t. \quad (13)$$

Therefore, the model can be casted as the following set of two equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \beta^{-1} E_t \{y_{t+1}\} / y_t \end{cases} \quad (14)$$

The first equation is the quantity of fruit falling from trees, and the second equation is the equilibrium condition on deposits. In absence of endogenous production, the number of fruit over time determines the equilibrium interest rate that ensure that ?? holds to equality.

Note that in this simple setup, the household does not optimally choose consumption, because it is exogenous (falling from trees). As a result, the consumption-saving is not really a saving problem because saving is zero in equilibrium. So the interpretation of the model is the following: the agent in t contemplates how much fruits y_t she has by observing ε_t (i.e. the weather). Given this new information, the agent plans how much fruits y_{t+1} he may expect at the next period. Given the current and expected levels of consumption, the equilibrium rate adjusts in order to clear financial markets, i.e. the real rate ensures that saving remains zero in equilibrium.

2.4 Expectations

How are we supposed to cope with the term $E_t \{y_{t+1}\}$ in the Euler equation from system (??)? This question is still left open in macroeconomics, as there is no perfect specification for the expectation scheme that portrays sufficiently well the formation process of expectations of agents over time.

Back in the 60s, macroeconomic analysis was relying on large-scale models based on old Keynesian economics. These models were 'backward looking', in a sense that they did not feature any type of expectation in their modeling structure. Some purely forward looking dynamics were simply omitted, which questioned their empirical relevance for policy analysis as argued by ? in his famous critique.

Which forward looking dynamics are we talking about? Let us think about some simple examples in macroeconomics:

1. A first example is about **price setting**. If the central bank is accommodative today, a worker may expect that inflation will rise in the future, so the worker asks today for an increase of her wage to compensate future inflation (which in turn self-fulfills the increase in inflation now). These considerations led to a New reinterpretation of the Keynesian Phillips curve.
2. Another example about **portfolio allocation in open economies**. Suppose that a central bank in the domestic country rises its interest rate, following this change, investing in the domestic country is more profitable, so you may expect that capital will flow to the domestic country, so you expect that your domestic currency will rise in the future. As a result, the domestic currency immediately rises to an increase in the domestic real rate prior the inflow of foreign capitals. This narrative describes the uncovered interest rate parity condition. These two examples are simply not incorporated in purely backward models.

Back to our problem of having an expectation scheme, Rational Expectations (RE) hypothesis has been the most popular tool to solve current macroeconomic model. RE provides a quick, simple and agnostic way of modelling expectations in macroeconomic models. In addition, RE are typically 'easy' to compute, as you even get closed-form expressions. The concept of rational expectations is pioneered by ?, an agent having rational expectation **use all relevant information** in forming expectations of economic variables. In our toy model, what is the relevant information for $E_t \{y_{t+1}\}$? Another implicit facet of RE lies in assumption that agents also have perfect knowledge of the model's equation. So if our agent knows the endowment in ??, by iterating forward this equation we get:

$$\begin{aligned} E_t \{y_{t+1}\} &= E_t \{1 - \rho + \rho y_t + \varepsilon_{t+1}\} \\ &= E_t \{1 - \rho + \rho y_t\} \\ &= 1 - \rho + \rho y_t, \end{aligned}$$

So as you can see, we are able to transform $E_t \{y_{t+1}\}$ into a system of current and backward variables. Implicitly, you may have noticed that we imposed $E_t \{\varepsilon_{t+1}\} = 0$. This is because the agent ignores the future realizations of the shock, but know its distribution. Since on average, $E(\varepsilon_t) = 0$, then it is rational to expect that – on average – the shocks are 0. So agent's expectations may be wrong, but **are correct on average** over time.

So, forecast errors are non-zero:

$$E_t \{y_{t+1}\} - y_{t+1} \neq 0 \text{ for } t > 0$$

But converge to zero asymptotically:

$$E \left(E_t \{y_{t+1}\} - y_{t+1} \right)_{t=0}^{+\infty} = 0$$

Back to our two-equation system in ??, under rational expectations, the model becomes:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases} . \quad (15)$$

Given this system, we are able to simulate our model because as long as we know the value of output, we can obtain the value of the real interest rate.

Exercise 1: Simulating the non-linear model

Consider the system in ??.

1. Write a MATLAB file that simulates this system over 400 periods. Use the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
2. Plot the simulation.
3. Compute the standard deviation of output and its correlation. Compare with the empirical data of real output that we have computed using a linear trend.
4. If you find a gap, adjust the calibration of σ and ρ to match US data.
5. Simulate an Impulse Response Function (IRF) of the model.
6. Does a large output expansion has the same effect on the real rate than a large depression? Compare the absolute values of the IRFs.
7. Plot the histograms of your simulated data and explain why the asymmetry is different for the distribution of output and the real rate.

3. Approximation methods

3.1 The problem

Our 2 equation model is a non-linear system with rational expectations. In compact notation, the system can be represented as follows:

$$\mathbb{E}_t [f_\theta (x_{t+1}, x_t, x_{t-1}, \epsilon_t)] = 0 \quad (16)$$

where:

- x is the vector collecting n endogenous variables;
- ϵ is the vector collecting the p exogenous stochastic shocks;
- θ is the vector collecting the deep parameters of the model;
- $f_\theta(\cdot)$ is the equations of the model;

The solution to a model like this is normally a set of functions relating controls to states such that our 2 equations are satisfied at all times (i.e. in $t+1$, t and $t-1$). The solution of the system x_t takes the form a decision rule:

$$x_t = g(x_{t-1}, \epsilon_t), \quad (17)$$

where function $g(\cdot)$ is unknown, but theoretically can be computed from ???. Let us first rewrite the system in ??? by injecting the model's solution in ???. Recall, that under rational expectations, $E_t x_{t+1} = g(x_t, 0_p)$ where $E(\epsilon_t) = 0_p$. To get rid off of the expectation term in the previous expression, we obtain:

$$f_\theta(g(x_t, 0_p), x_t, y_{t-1}, \epsilon_t) = 0. \quad (18)$$

Iterating again to express our system as a function of x_{t-1} :

$$f_\theta(g(g(x_{t-1}, \epsilon_t), 0_p), g(x_{t-1}, \epsilon_t), x_{t-1}, \epsilon_t) = 0 \quad (19)$$

Therefore, the true solution of the system ??? is a function $g(\cdot)$ that satisfies ???. The problem is that there is an infinite number of values of y_{t-1} and ϵ_t and an infinite number of values of y_t to find. Hence, with a few rare exceptions, what we do to solve DSGE models is to approximate $g(\cdot)$. Note that in our two-dimensional model of consumption-saving, we are able to get a closed-form expression $g(\cdot)$ but in more complicated models, it's impossible to obtain the true value of $g(\cdot)$. The next subsection aims at explaining how we can approximate $g(\cdot)$.

The closed-form expression of the policy rule is given by:

$$y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \quad (20)$$

$$r_t = \frac{\rho}{\beta} + \frac{\beta^{-1}(1 - \rho)}{1 - \rho + \rho y_{t-1} + \varepsilon_t}. \quad (21)$$

3.2 Linearization Method

The standard way to bound the problem is to approximate the function $v(\cdot)$ by a log-linear function around the steady state using the method of ?. Under particular conditions defined by ?, our linear rational expectations model has a stable and convergent intertemporal equilibrium.

More precisely, we usually take the following steps to solve a model:

1. Find the steady state of the model;
2. (log-)Linearize around the steady state;
3. Write the Dynare file.

3.3 Linearization application

For any function $f(x_t)$ with corresponding steady state $f(\bar{x})$, a Taylor expansion up to first order is given by:

$$f(x_t) \simeq f(\bar{x}) + f'(\bar{x})(x_t - \bar{x}) \quad (22)$$

Expressed in terms of deviation from the steady state, $f(x_t) - f(\bar{x})$, the expansion can be rewritten in the following way:

$$\begin{aligned} f(x_t) - f(\bar{x}) &\simeq f'(\bar{x})(x_t - \bar{x}) \\ &\simeq f'(\bar{x})\bar{x} \frac{(x_t - \bar{x})}{\bar{x}} \\ &\simeq f'(\bar{x})\bar{x} \log\left(\frac{x_t}{\bar{x}}\right) \\ &\simeq f'(\bar{x})\bar{x} \hat{x}_t \end{aligned}$$

where $\hat{x}_t = \log(x_t/\bar{x})$ is the percentage variations of x_t around its steady state.

For a two-variable function, the approximation reads as:

$$f(x_t, y_t) - f(\bar{x}, \bar{y}) \simeq f'_x(\bar{x}, \bar{y})\bar{x}\hat{x}_t + f'_y(\bar{x}, \bar{y})\bar{y}\hat{y}_t \quad (23)$$

We can then summarize these general approximations by some tricks that we can use to log-linearize quickly a model:

Level variable	Log-linearized variable
x_t	$\bar{x}\hat{x}_t$
$x_t y_t$	$\bar{x}\bar{y}(\hat{x}_t + \hat{y}_t)$
$\frac{x_t}{y_t}$	$\frac{\bar{x}}{\bar{y}}(\hat{x}_t - \hat{y}_t)$
x_t^a	$\bar{x}^a a \hat{x}_t$
$\log(x_t)$	$\hat{x}_t$
$\exp(x_t)$	$\exp(\bar{x})\bar{x}\hat{x}_t$

3.3 Linearization application

Let us now linearize our system of equations:

$$\begin{cases} y_t = 1 - \rho + \rho y_{t-1} + \varepsilon_t \\ r_t = \frac{(1-\rho+\rho y_t)}{\beta y_t} \end{cases} \quad (24)$$

Steady state. A preliminary step before performing the approximation to our model's equations is to characterize the steady state of the model (i.e. the fixed point of the approximation). Following the linearization, each variable is expressed in percentage deviation from the steady state. To obtain the steady state, one just needs to take off the time subscript on endogenous variables:

$$\begin{cases} \bar{y} = 1 - \rho + \rho \bar{y} \\ \bar{r} = \frac{(1-\rho+\rho \bar{y})}{\beta \bar{y}} \end{cases} \quad (25)$$

Note that the following system in the (deterministic) steady state reads as follows:

$$\begin{cases} \bar{y} = 1 \\ \bar{r} = \frac{1}{\beta} \end{cases} \quad (26)$$

Linearizing the equations. Next, we must linearize the equations of the model. We apply here:

$$\underbrace{y_t}_{\bar{y}\hat{y}_t} = \underbrace{1 - \rho}_0 + \underbrace{\rho y_{t-1}}_{\rho\bar{y}\hat{y}_{t-1}} + \underbrace{\varepsilon_t}_{\varepsilon_t}$$

Note that since $\bar{y} = 1$, then equation boils down to:

$$\hat{y}_t \simeq \rho\hat{y}_{t-1} + \varepsilon_t$$

Some remarks on the linearization. You may struggle to log-linearize, but with learning-by-doing, you will get used to it. After some time of practice, linearizing equations get very quickly. Note also that the stochastic shock is not affected by the linearization process

$$\begin{aligned} r_t &= \frac{(1 - \rho + \rho y_t)}{\beta y_t} \\ r_t &= \frac{(1 - \rho)}{\beta} \frac{1}{y_t} + \frac{\rho}{\beta} \\ \bar{r}\hat{r}_t &\simeq -\frac{1 - \rho}{\beta\bar{y}}\hat{y}_t \\ \hat{r}_t &\simeq -(1 - \rho)\hat{y}_t \end{aligned}$$

Exercise 2: Simulating the linear model

Consider now the linear system.

1. Write a MATLAB file that simulates this system over 400 periods. Use the following calibration: $\beta = 0.99$, $\sigma = 0.1$ and $\rho = 0.85$.
2. Plot the simulation.
3. Simulate an Impulse Response Function (IRF) of the model.
4. Does a large output expansion has the same effect on the real rate than a large depression? Compare the absolute values of the IRFs.
5. Plot the histograms of your simulated data and explain why the asymmetry is different for the distribution of output and the real rate.
6. Compute the Errors of approximation on the nominal rate $E[(r_t - \hat{r}_t)^2]$.

4. A production economy

Let us now suppose that production is endogenous in the same spirit as ?. The welfare index of the representative household reads as follows:

$$\mathcal{W}_t = \mathbb{E}_t \sum_{\tau=0}^{+\infty} \theta_{t+\tau} \mathcal{U}(C_{j,t+\tau}) \quad (27)$$

Note in this specific case here, we deal with an unit root component. To takeaway the unit root in the data generating process of consumption, we assume as ? that the stochastic discount factor is time varying:

$$\theta_{t+\tau} = \begin{cases} \beta^\tau \exp(-\chi [C_{t-1+\tau} - \bar{C}] / \bar{C}), & \text{if } \tau > 0 \\ 1 & \text{else} \end{cases} \quad (28)$$

where $\bar{C}$ is the steady state consumption and β the discount factor. Note that θ_t is economy-wide, so the agent j does not optimize over C_t . Note that in our model, we only need $\theta_t = \beta$ and $\theta_{t+1} = \beta \exp(-\chi [C_{t+1} - \bar{C}] / \bar{C})$.

The households faces a (intertemporal) budget constraint:

$$C_{jt} + I_{jt} = Y_{jt} \quad (29)$$

The total income Y_{jt} is spent on consumption goods C_{jt} and investment goods I_{jt} . An intertemporal budget constraint shows what consumption options are available between the present and future. The constraint can be used to determine the future value of present income or the present value of future income. This constraint binds each period, *i.e.* households expenditures equal her income for any value of t . Here, the budget constraint can also be interpreted as the definition of the GDP using expenditure approach (neglecting both stocks and the trade balance).

The law of motion of capital reads as in the ?'s model:

$$I_{jt} = K_{jt} - (1 - \delta) K_{j,t-1} \quad (30)$$

Investment is determined by the gap between current capital K_{jt} and depreciated capital $(1 - \delta) K_{j,t-1}$. In the latter expression, the parameter $0 \leq \delta \leq 1$ denotes the depreciation rate of capital.

The standard Cobb-Douglas technology is determined by:

$$Y_{jt} = A_t K_{j,t-1} \quad (31)$$

where the production process of new goods is denoted Y_{jt} while A_t is a productivity shock which temporary affects the society's ability to produce goods, it affects all the households in our economy in the same way. The production combines technology and capital $K_{j,t-1}$ bought in $t - 1$ as capital requires one period to be settled (the so called time-to-build in the RBC literature). Concerning the productivity shock, it is assumed to follow a simple AR(1) process given by:

$$\log(A_t) = \rho \log(A_{t-1}) + \eta_t, \quad (32)$$

where $0 \leq \rho < 1$ is the autocorrelation coefficient of the shock and η_t is an innovation distributed normally with zero mean and finite variance σ^2 such that $\eta_t \sim \mathcal{N}(0, \sigma^2)$.

To summarize, the system reads as follows:

$$\begin{aligned} \max_{C_{jt}, K_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \theta_{t+\tau} \log(C_{j,t+\tau}) \\ \text{s.t. : } C_{jt} + K_{jt} = A_t K_{j,t-1} + (1 - \delta) K_{j,t-1} \end{aligned}$$

Based on this model, we will work on the following exercises.

Exercise 3: A production economy

Consider the production economy model.

1. Set the Lagrangian problem.
2. Find the first order conditions. Get rid off the Lagrangian multiplier to obtain the Euler equation on physical capital.
3. Write the three equations of the model. Find the steady state.
4. Linearize the three equations. Note that the steady state of the discount factor is $\bar{\theta} = \beta$, while the approximation of the discount factor reads as follows $\theta_{t+1} \simeq \beta (1 - \chi \hat{c}_t)$.
5. Simulate your model in dynare with $\beta = 0.99$, $\sigma = 0.01$, $\rho = 0.95$, $\delta = 0.025$ and $\chi = 0.01$.
6. Imposing $\chi = 0$ allows the presence of a unit root in the model. How can we detect the unit root in the eigen values and in the moments of the model?

Exercise 4: extending the simple model

1. Let us now assume decreasing return to scale in the production function: $Y_{jt} = A_t K_{jt-1}^\alpha$ with α the capital intensity parameter. We impose $\alpha = 0.40$. Note that negative scale returns remove the unit root experiments, so we don't need θ_t anymore and thus assume $\theta_t = \beta$.
 - (a) Update the first order conditions.
 - (b) Find steady state.
 - (c) Log-linearize.
 - (d) Update your dynare file.
2. The GDP, Y_t and investment I_t are not (yet) included in the model. Include them and draw their responses to a productivity shock. To do so, log-linearize their equations and find their steady state.
3. Check whether the investment- to-gdp ratio $\bar{I}/\bar{Y}$ and $\bar{C}/\bar{Y}$ are consistent with the data (on average, investment and consumption represent respectively 20% and 60% of the GDP for OECD countries). Why is there such a huge gap? How could you improve the RBC model?
4. Change the calibration to get a non-stationnary system response. What is Dynare telling you in the command window? Does increasing return to scale makes the model non-stationnary?

Exercise 5: extension: habits in consumption

Theoretical macro models typically failed at reproducing the excess return premium on financial markets before the 90s. In asset pricing theory, a new consensus emerged to solve these puzzle through the inclusion of new ingredients in macro-models which would help in solving the puzzle. One of them is the consumption habits, pioneered by ? and ?. Typically in real life situations, households have strong consumption habits which materializes in time series by having a very stable consumption over the business cycles, compared to other aggregates such as investment and output. Basically, if the household decides to consume more, it will raise the amount of consumption tomorrow. However, there is few evidence about such behaviour at a micro-level, suggesting that consumption habits is probably a shortcut for a more complex explanation of consumption inertia.

1. Let us assume first *external* habits. These external habits simply change how households are enjoying consumption in their utility function:

$$\mathcal{U}(C_{jt} - hC_{t-1}) \quad (33)$$

where $h \in [0, 1)$ is the degree of consumption habits. Here, it is critical to notice that these habits are depending on the aggregate amount of consumption C_{t-1} (i.e. there is no subscript j). So basically, the amount of aggregate consumption is not a control variable when households maximizes utility. This is referred to as *catching up with the joneses* as people will copy the behaviour of the majority of consumers.^a

- (a) The marginal utility of consumption reads as : $\lambda_t = \mathcal{U}'(C_{jt} - hC_{t-1})$.
 - (b) Find the steady state.
 - (c) Linearize the expression. It might be useful to split the problem with $\lambda_t = \mathcal{U}'(x_t)$ and $x_t = C_t - hC_{t-1}$ because the linearization trick cannot be applied for highly non-linear problems.
 - (d) How persistent is now consumption?
2. Let us now assume internal habits, denoted $\mathcal{U}(C_{jt} - hC_{jt-1})$.
 - (a) The marginal utility of consumption reads as : $\lambda_t = \mathcal{U}'(C_{jt} - hC_{jt-1}) - h\beta\mathcal{U}'(C_{jt+1} - hC_{jt})$.
 - (b) Find the steady state.
 - (c) Linearize the expression. It might be useful to split again the problem with $\lambda_t = \mathcal{U}'(x_t) - \beta h\mathcal{U}'(x_{t+1})$ and $x_t = C_t - hC_{t-1}$.
 - (d) How persistent is now consumption?

^aSee ? the story about the Joneses.

Exercise 6: demand shocks

Assume now that our economy has a government. The government finances public spending by charging a lump-sum tax T_{jt} on households. The total amount of public spending reads as, $P_t G_t$. The balance sheet of government writes:

$$P_t G_t = \int_0^1 T_{jt} dj \quad (=T_t). \quad (34)$$

Since the economy is populated by an uniform household, we can write T_t directly. The budget constraint of the representative household is now:

$$C_{jt} + I_{jt} = A_t K_{jt-1}^\alpha - \frac{T_{jt}}{P_t}, \quad (35)$$

while the resource constraint/the GDP through the expenditure approach is now given by:

$$Y_t = C_t + I_t + G_t. \quad (36)$$

The decision of increasing/decreasing spendings is decided exogenously by the government. In absence of an explicit modelling of political cycles, the spending is assumed to vary around a fixed value of output:

$$G_t = \bar{G} \varepsilon_t^G = g^y \bar{Y} \varepsilon_t^G, \quad (37)$$

where spending follows an $AR(1)$ process:

$$\log(\varepsilon_t^G) = \rho_G \log(\varepsilon_{t-1}^G) + \eta_t^G, \quad (38)$$

with $0 \leq \rho_G < 1$ is the autocorrelation coefficient of the demand shock and η_t^G is an innovation distributed normally with zero mean and finite variance σ_G^2 such that $\eta_t^G \sim \mathcal{N}(0, \sigma_G^2)$. Parameter g^y is the spending-to-GDP ratio. We borrow the values of ? and assume that $\rho_G = 0.97$, $\sigma_G = 0.053$ and $g^y = 0.20$. Implicitly in steady state, the shock is normalized to one $\bar{\varepsilon}^G = 1$.

1. Compute the new steady state of the model. *Tips:* (i) define a new parameter, $g^y = 0.2$, which affects the new steady state; (ii) since the budget is balanced, you can substitute real taxes T_t/P_t by spending G_t .
2. Let's now calibrate the demand shock process, show the IRF after the realization of the demand shock.
3. According to Keynesian economics, is this shock consistent with this theory? Why do households are "Ricardian"? Is this behaviour observed in the data?
4. Analyse the variance decomposition, what is the main driver of output?
5. Is this RBC model now able to replicate another business cycle fact which says output is more volatile than consumption?

We consider a time-preference shock which is also considered as a demand shock, *i.e.* the household discount factor β is affected by an exogenous component ε_t^C . Basically, the representative household temporary offsets her consumption smoothing by increasing (decreasing) its spending

for a positive (negative) realization of this time-preference shock, as she becomes transtory more patient (impatient). The new objective function of the household reads as:

$$\max_{\{C_{jt}, K_{jt}\}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^\tau \varepsilon_{t+\tau}^C \log(C_{jt+\tau}) \quad (39)$$

where the time preference shock reads as:

$$\log(\varepsilon_t^C) = \rho_C \log(\varepsilon_{t-1}^C) + \eta_t^C. \quad (40)$$

We assume the following calibration as in ? for this shock: $\bar{\varepsilon}^C = 1$, $\rho_C = 0.22$ and $\sigma_C = 0.023$.

6. How is the Euler equation affected by this shock? Take it in logs.
7. Simulate the system response, what is the effect of the shock on the economy? Analyse the variance decomposition.
- 8 Using simultaneously all the shock, analyse the forecast error variance decomposition of the model up to 1, 2, 4, 10 and 100 quarters (Find the function in Dynare's userguide). Which shocks is driving output in the short and in the long run?